

Winning Space Race with Data Science

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https://github.com/nicholas-yap/spacex-data-science-capstone/

Outline

Executive summary

Model evaluation, confusion matrix and selection

Conclusions

Project background and methods

EDA, SQL, maps, dashboard and sources

Executive Summary

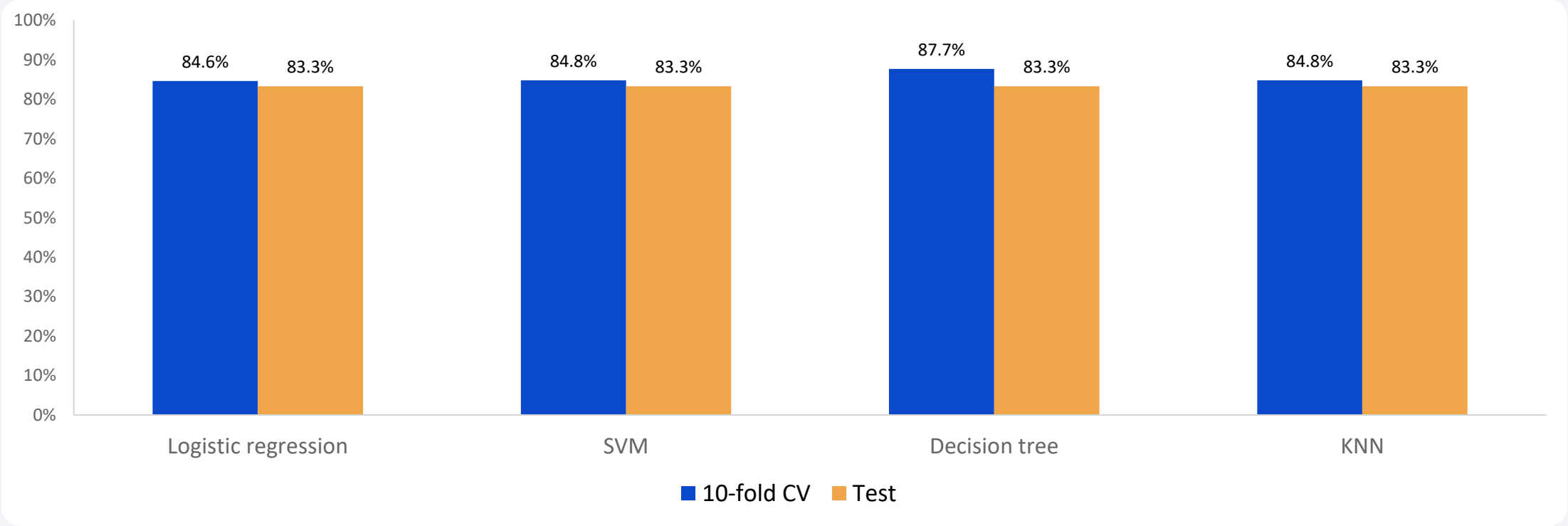
90 Falcon 9 launches form the modeling sample. The course label marks 60 successes (66.7%).

KSC LC 39A has the highest observed site rate: 77.3% (17/22). CCAFS SLC 40 contributes 33 of 60 successes.

All four models score 83.33% on 18 test launches. The decision tree leads cross-validation at 87.68%.

Small samples, changing technology and preprocessing leakage limit generalization.

Classification Accuracy



All models tie at 83.33% on the test set. The decision tree leads CV at 87.68%. The always-Class-1 test baseline is 66.67%.

Decision Tree Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	3	3
Class 1	0	12

15/18 correct (83.33%)
Class 1 precision: 12/15 = 80%
Class 1 recall: 12/12 = 100%

3 of 6 actual Class 0 outcomes are false positives

Model Evaluation and Selection

Classifier	10-fold CV accuracy	Test accuracy
Logistic regression	84.64%	83.33%
Support vector machine	84.82%	83.33%
Decision tree	87.68%	83.33%
K-nearest neighbors	84.82%	83.33%

Decision tree is selected for its highest cross-validation accuracy: 87.68%. All four models tie at 83.33% on 18 test launches.

Split: 72 train, 18 test. Small samples and preprocessing leakage limit confidence.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

Conclusions

Successful first-stage outcomes become more common over the study period, but progress is uneven across years.

KSC has the highest observed site rate. CCAFS contributes the most successes. Orbit and payload patterns depend on mission mix.

Four classifiers tie on the small test set. The tree's CV lead does not establish better future performance.

Next evaluation: fit preprocessing within training folds, remove features unavailable before launch, and use a later time period for testing.

Introduction

Business question: how can historical launch characteristics help estimate the chance of a successful first-stage outcome?

Reusable boosters create a reason to study landing outcomes. This project predicts the course outcome label, not launch price or cost savings.

Compare launch sites, payload, orbit and experience, then evaluate four classification models.

Scope: historical course snapshots. A successful mission and a successful first-stage outcome measure different events.

Section 1

Methodology



Methodology

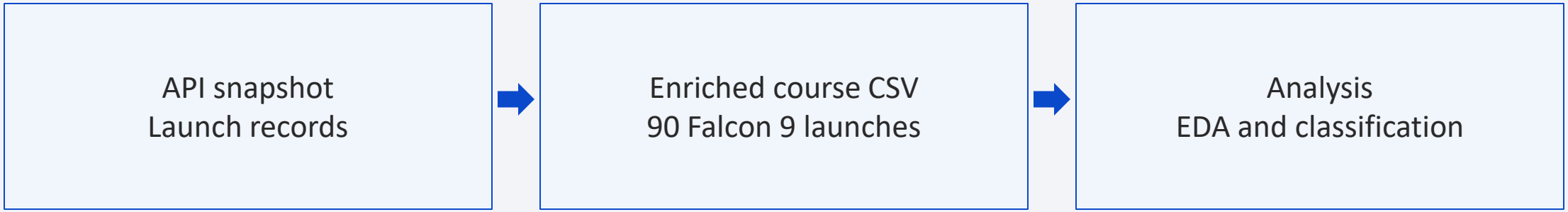
Collect historical launch records from a SpaceX API snapshot and a Wikipedia launch-table snapshot.

Clean outcomes and missing payloads. Use 90 launch records and 83 encoded features for classification.

Explore relationships with charts and SQLite queries. Add Folium maps and a Dash dashboard.

Compare tuned classifiers using 10-fold cross-validation and the saved 18-launch test split.

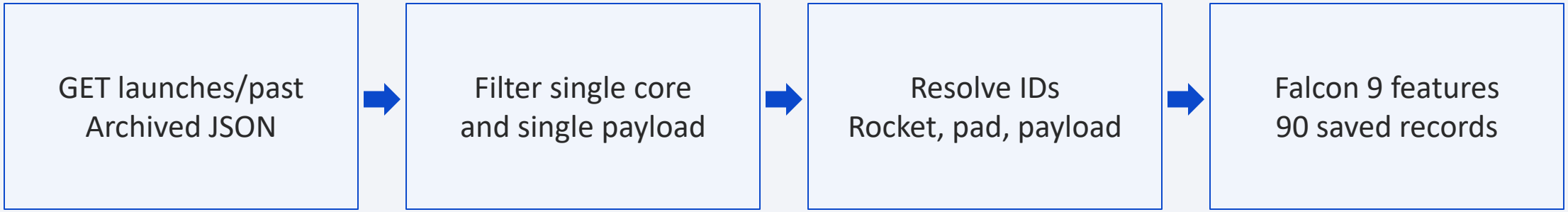
Data Collection



A separate Wikipedia scrape contains 121 launches. The SQL snapshot contains 101 records.

These sources have different coverage and labels. Results retain their source-specific denominators.

Data Collection – SpaceX API

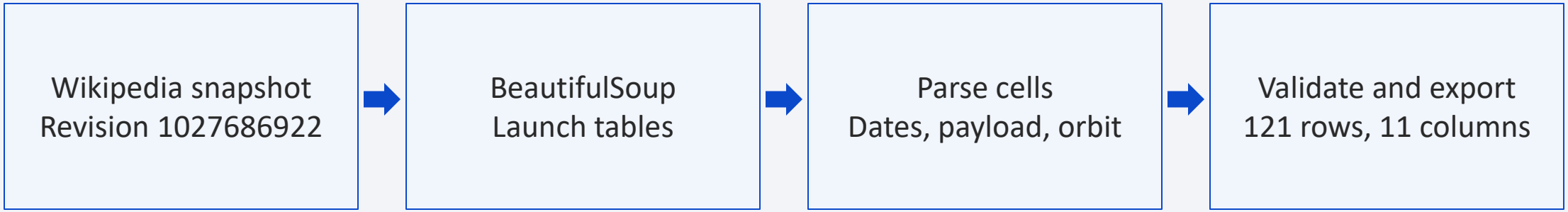


The archived launch response supports reproducible inspection. The supplied course CSV contains the enriched rocket, launchpad, payload and core fields.

The companion notebook validates the snapshot and the 90-row feature table without claiming a fresh API extraction.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/01-data-collection.ipynb

Data Collection – Web Scraping

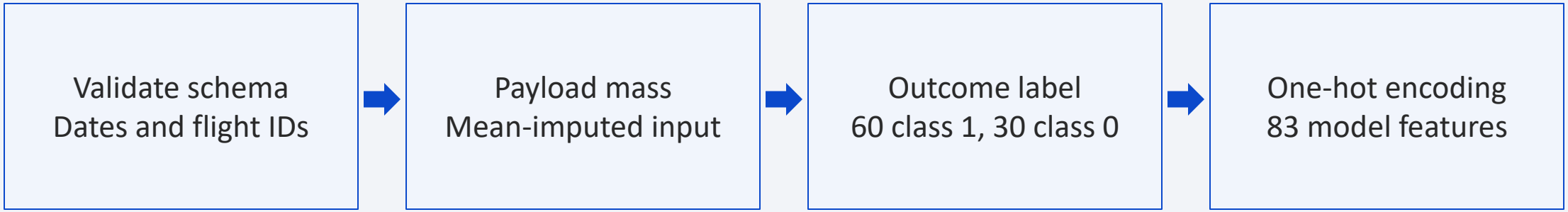


The completed notebook extracts numbered launch rows and checks flight IDs, dates and times.

The June 2021 page snapshot extends beyond the 90-row modeling sample. The scraped table remains a separate source.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/jupyter-labs-webscraping.ipynb

Data Wrangling



Class 1 includes True ASDS, True RTLS and True Ocean. Class 0 includes failed, missing and no-attempt outcomes.

Controlled ocean outcomes do not establish booster recovery. Keep this distinction when interpreting “success.”

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/02-wrangling-eda.ipynb

EDA with Data Visualization

Scatter plots compare flight number and payload with launch site and orbit. Color separates the two outcome classes.

Orbit success bars show rates with launch counts so small groups remain visible.

The yearly line chart shows changes in outcome rate, with year-specific denominators.

Charts describe associations. Site, orbit, payload and technology changed together over time.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/02-wrangling-eda.ipynb

EDA with SQL

DISTINCT and LIKE identify launch-site labels and the first five CCA records.

SUM, AVG and MAX summarize payload mass and the boosters carrying the maximum.

Date and outcome filters identify the first ground landing, drone ship landings and failures in 2015.

GROUP BY and COUNT summarize mission results and rank landing outcomes for the specified period.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Interactive Map with Folium

Site circles and labels locate the three modeled launch facilities on a continental map.

A MarkerCluster groups overlapping launches. Green marks Class 1 and red marks Class 0.

Popup details show flight, date, payload and outcome. Expanding a cluster separates markers for inspection.

Lines connect CCAFS SLC 40 to selected road, rail and coastline points. Haversine distance gives approximate straight-line separation.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/03-interactive-maps.ipynb

Dashboard with Plotly Dash

The site dropdown switches between all launches and a selected facility.

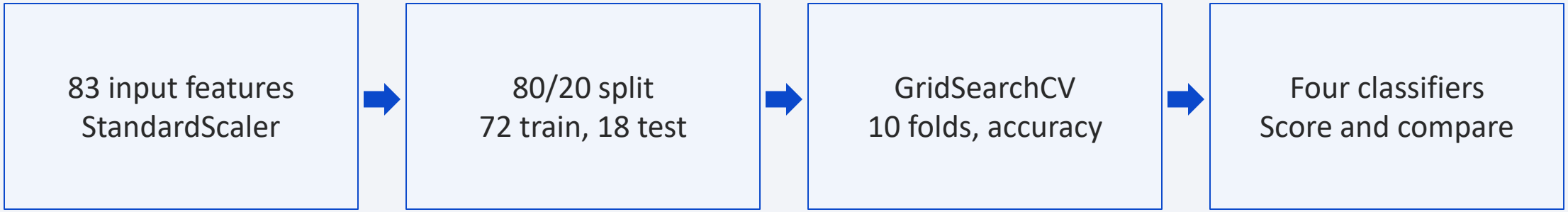
The all-sites pie chart shows each site’s contribution to successful outcomes. A selected-site pie shows its success ratio.

The payload range slider filters the outcome scatter plot. Color identifies booster block.

Hover details reveal launch date, flight number and outcome. The dashboard uses the same 90-row sample as the EDA.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/spacex_dashboard.py

Predictive Analysis



Logistic regression, SVM, decision tree and KNN use the saved lab configuration. random_state=2 fixes the train/test split.

The lab scales all 90 rows before splitting. Future evaluation should fit preprocessing inside each training fold and use a chronological holdout.

https://github.com/nicholas-yap/spacex-data-science-capstone/blob/main/notebooks/SpaceX-Machine-Learning-Prediction-Part-5-v1.ipynb

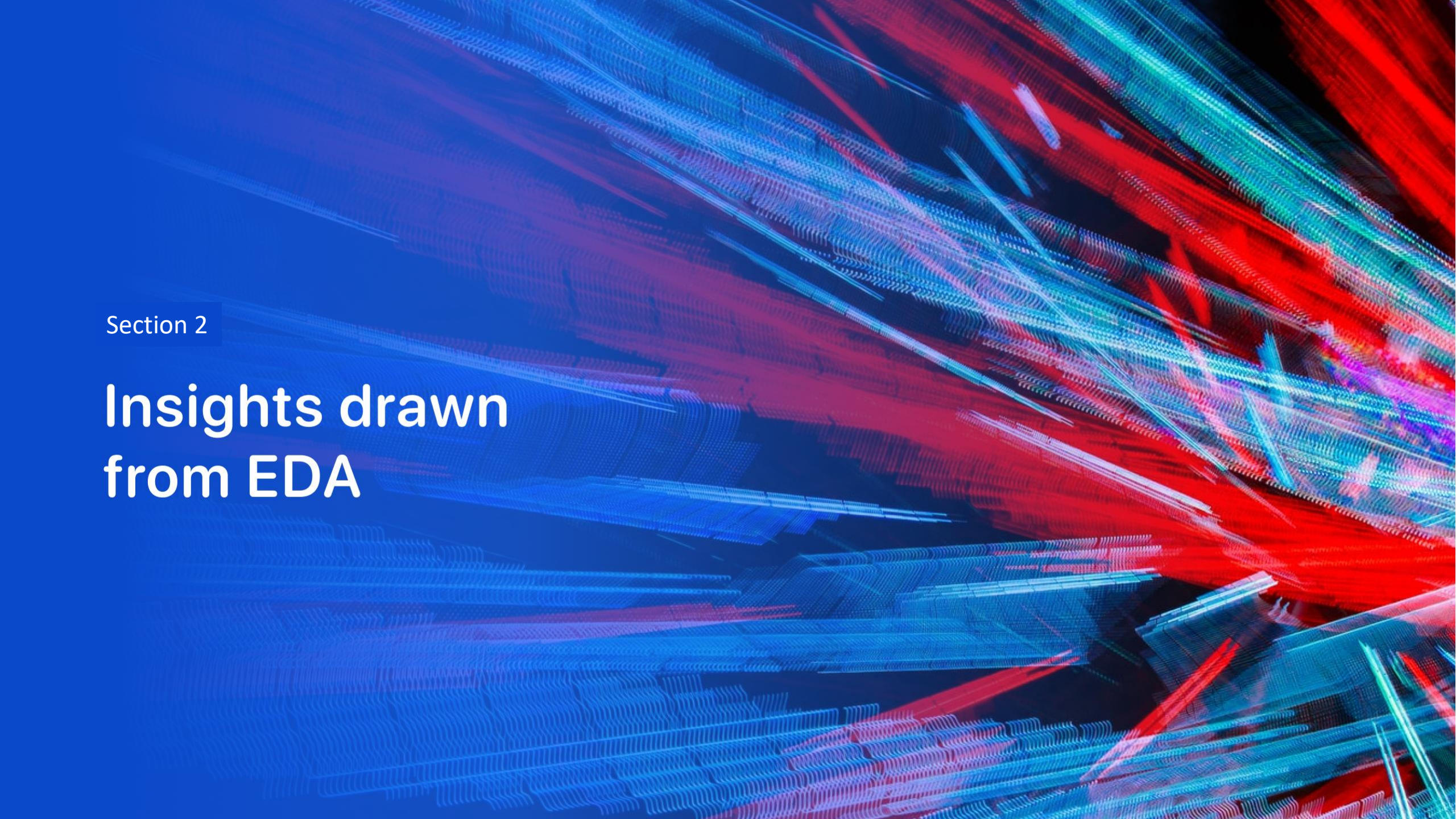
Results

EDA: outcome rates differ across sites, orbit groups and launch years.

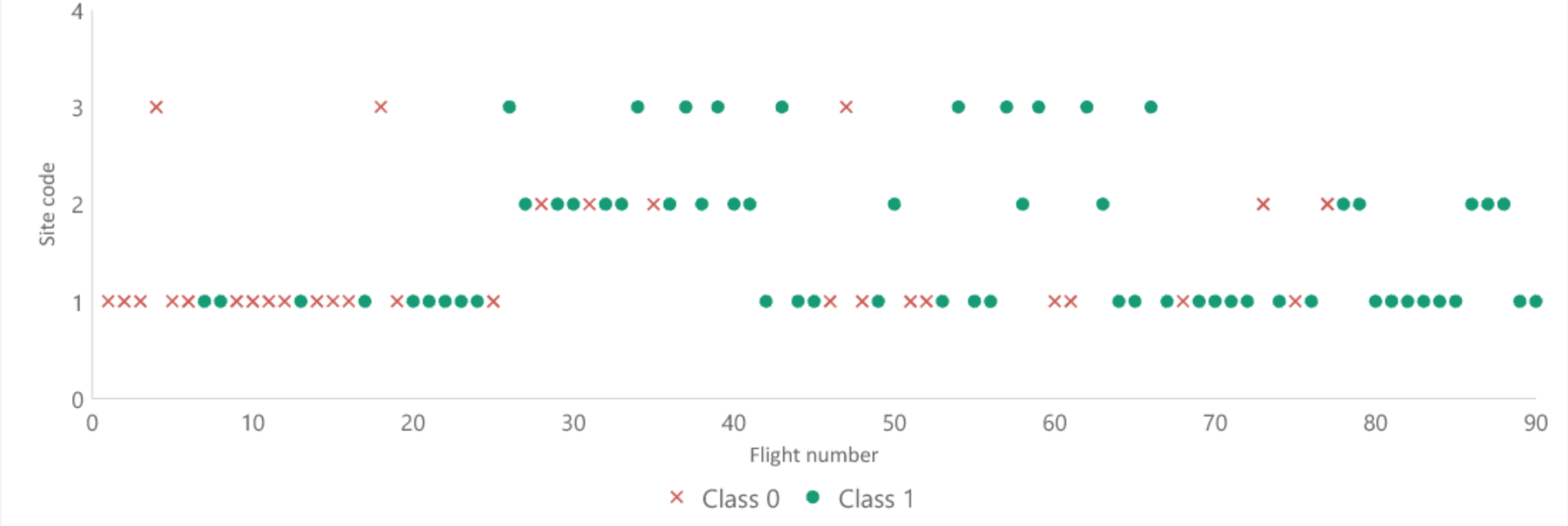
SQL: 101 missions include 100 mission successes and one failure. NASA (CRS) payload totals 45,596 kg.

Interactive views: coastal launch sites and payload filters help inspect individual records.

Prediction: each model correctly classifies 15 of 18 test launches. The test scores do not establish a unique winner.


Section 2
Insights drawn from EDA

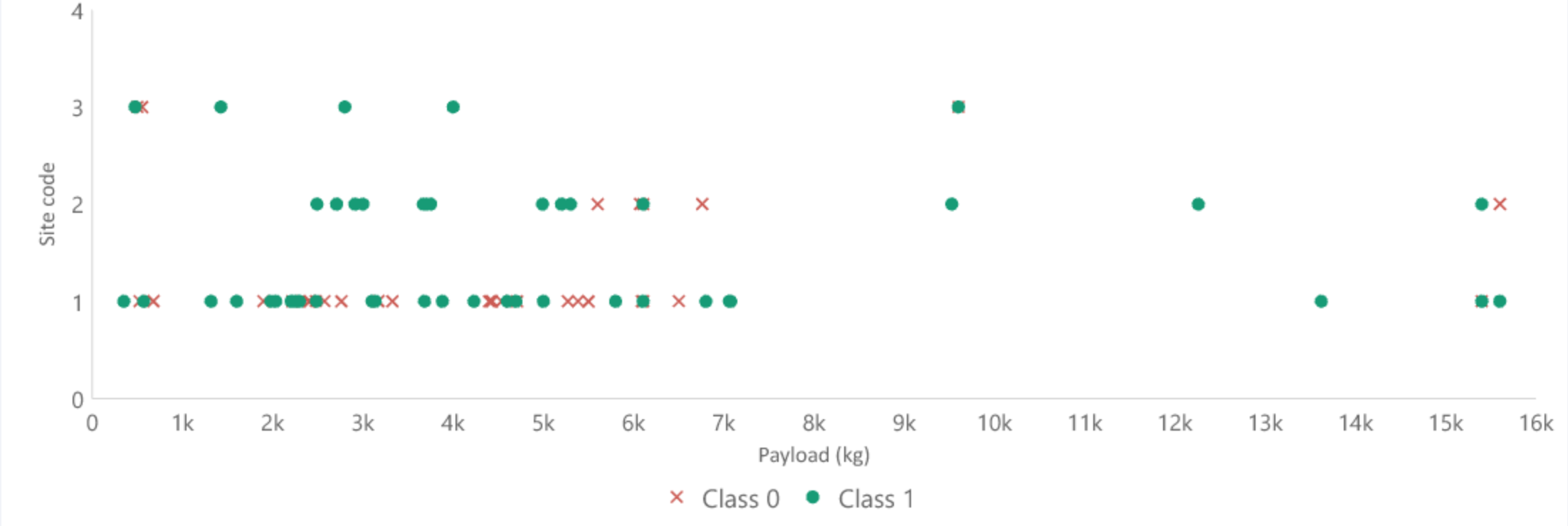
Flight Number vs. Launch Site



1: CCAFS SLC 40 2: KSC LC 39A 3: VAFB SLC 4E

Later launches show more Class 1 outcomes. Site and flight sequence overlap, so the plot cannot isolate a site effect.

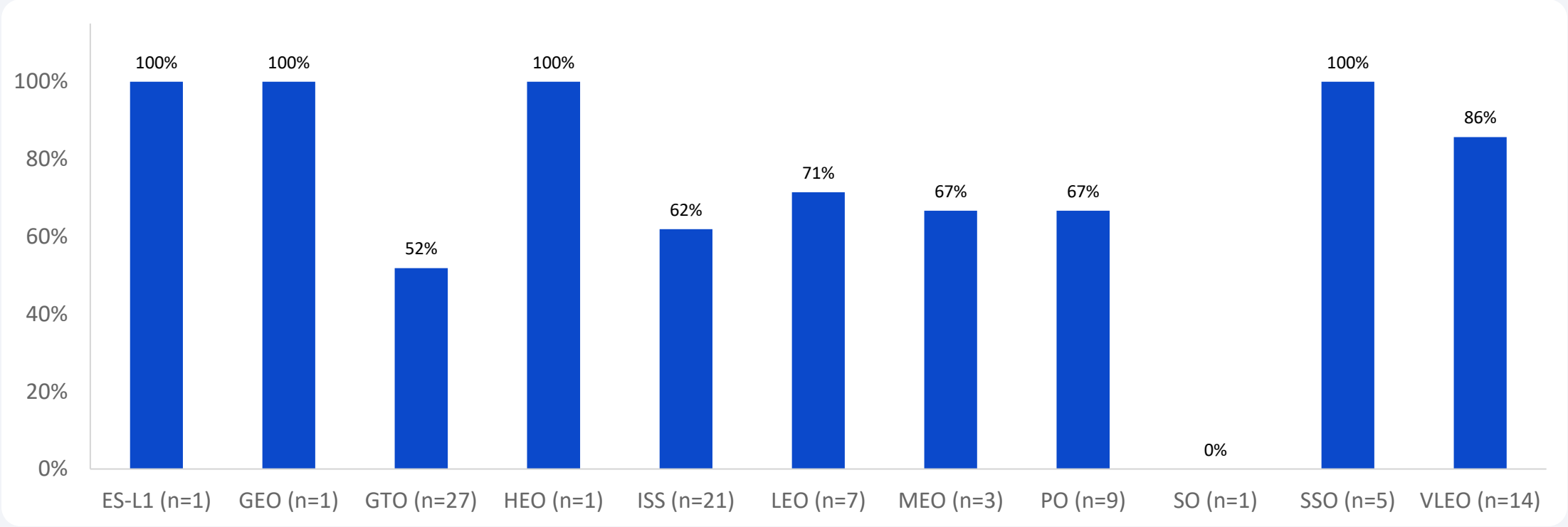
Payload vs. Launch Site



1: CCAFS SLC 40 2: KSC LC 39A 3: VAFB SLC 4E

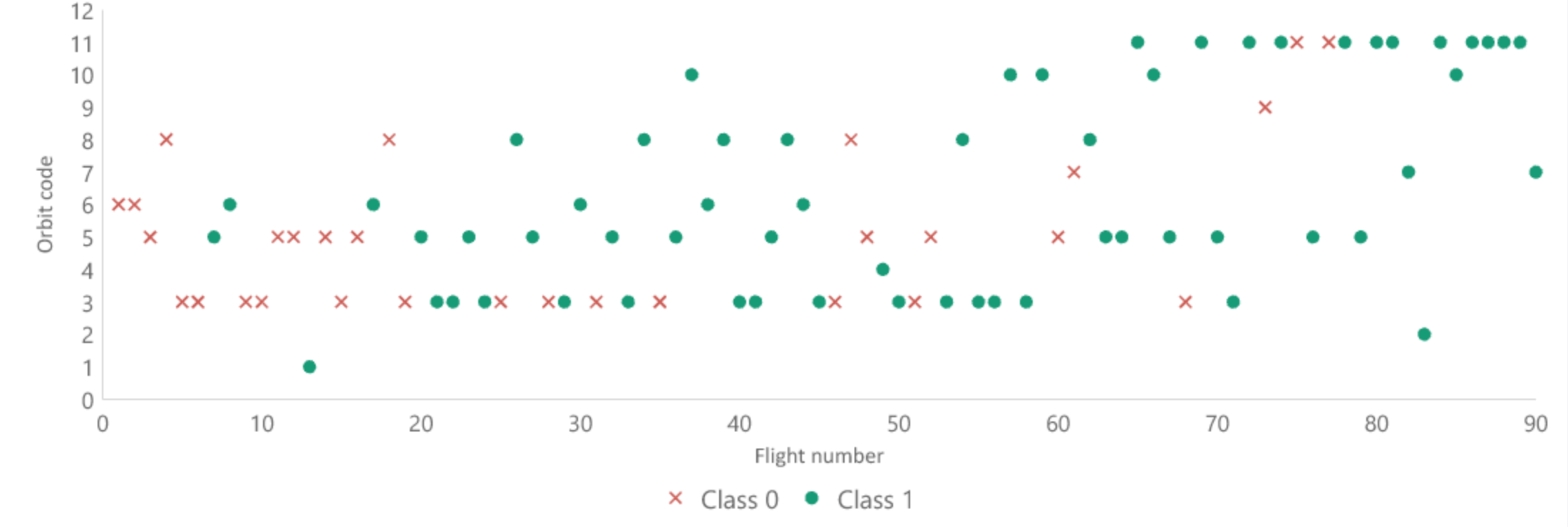
CCAFS SLC 40 covers the broadest history. Both outcome classes occur across payload ranges, and some payload values are mean-imputed.

Success Rate vs. Orbit Type



SSO records 5/5 successes. GEO, HEO and ES-L1 each have only one launch, so their 100% rates provide little evidence.

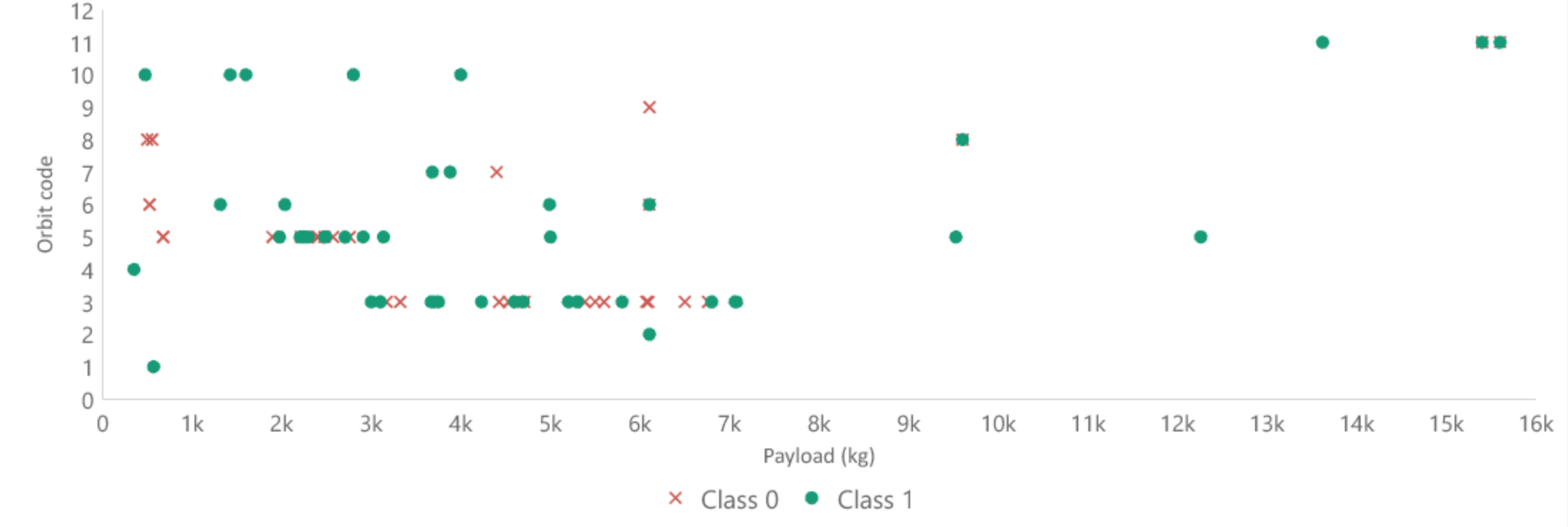
Flight Number vs. Orbit Type



1: ES-L1 2: GEO 3: GTO 4: HEO 5: ISS 6: LEO 7: MEO 8: PO 9: SO 10: SSO 11: VLEO

VLEO launches cluster late in the sample. Differences by orbit may partly reflect changes in booster technology and experience.

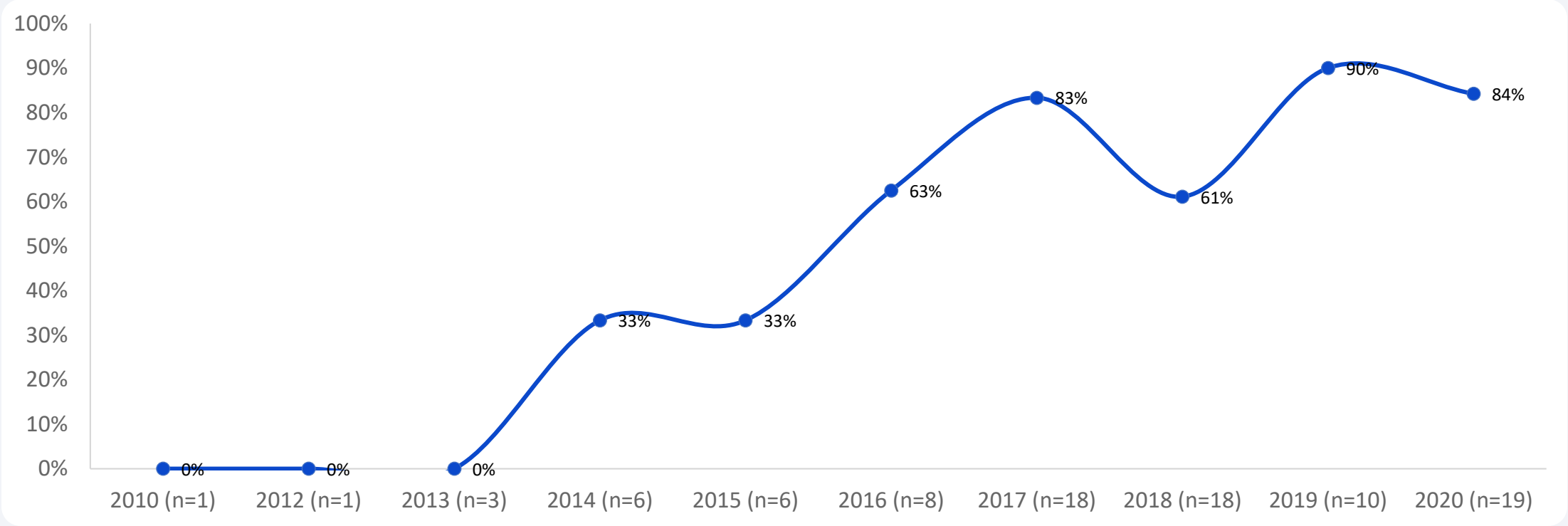
Payload vs. Orbit Type



1: ES-L1 2: GEO 3: GTO 4: HEO 5: ISS 6: LEO 7: MEO 8: PO 9: SO 10: SSO 11: VLEO

GTO has 14/27 Class 1 outcomes (51.9%). VLEO has 12/14 (85.7%). Payload alone does not explain the difference.

Yearly Outcome Trend



The rate rises from 0% in early years to 84.2% in 2020, with a dip in 2018. The sample contains no 2011 launch.

All Launch Site Names

Launch site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

The SQL source contains four labels. CCAFS LC-40 and CCAFS SLC-40 appear separately in this snapshot.

Launch Sites Beginning with ‘CCA’

Date	Booster version	Payload (kg)	Site
2010-06-04	F9 v1.0 B0003	0	CCAFS LC-40
2010-12-08	F9 v1.0 B0004	0	CCAFS LC-40
2012-05-22	F9 v1.0 B0005	525	CCAFS LC-40
2012-10-08	F9 v1.0 B0006	500	CCAFS LC-40
2013-03-01	F9 v1.0 B0007	677	CCAFS LC-40

The earliest five CCA records span June 2010 to March 2013. The query orders by date and UTC time before LIMIT 5.

Total NASA (CRS) Payload Mass

45,596 kg

SUM(PAYLOAD_MASS__KG_)
WHERE Customer = 'NASA (CRS)'

This is the NASA Commercial Resupply Services subset. It does not include every customer label containing NASA.

Average Payload Mass for F9 v1.1

2,928.4 kg

AVG(PAYLOAD_MASS__KG_)
WHERE Booster_Version = 'F9 v1.1'

The exact version label matches five rows. Including every label beginning with F9 v1.1 would answer a different query.

First Successful Ground Landing

22 December 2015

MIN(Date)
WHERE TRIM(Landing_Outcome) = 'Success (ground pad)'

This is the earliest successful ground-pad landing recorded in the SQL snapshot.

Drone Ship Landings at 4,000–6,000 kg

Booster version
F9 FT B1021.2
F9 FT B1031.2
F9 FT B1022
F9 FT B1026

Four boosters meet both conditions: successful drone ship landing and payload strictly above 4,000 kg and below 6,000 kg.

Mission Outcomes

Mission result	Missions
Success	100
Failure	1

100/101 missions count as successful. “Success (payload status unclear)” is included. Mission results differ from landing outcomes.

Boosters Carrying the Maximum Payload

Booster	Payload (kg)	Booster	Payload (kg)
F9 B5 B1048.4	15600	F9 B5 B1051.4	15600
F9 B5 B1048.5	15600	F9 B5 B1051.6	15600
F9 B5 B1049.4	15600	F9 B5 B1056.4	15600
F9 B5 B1049.5	15600	F9 B5 B1058.3	15600
F9 B5 B1049.7	15600	F9 B5 B1060.2	15600
F9 B5 B1051.3	15600	F9 B5 B1060.3	15600

Twelve booster-flight labels share the maximum payload of 15,600 kg in this snapshot.

Drone Ship Landing Failures in 2015

Month	Landing outcome	Booster	Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Two 2015 records show failed drone ship landings: January (B1012) and April (B1015), both at CCAFS LC-40.

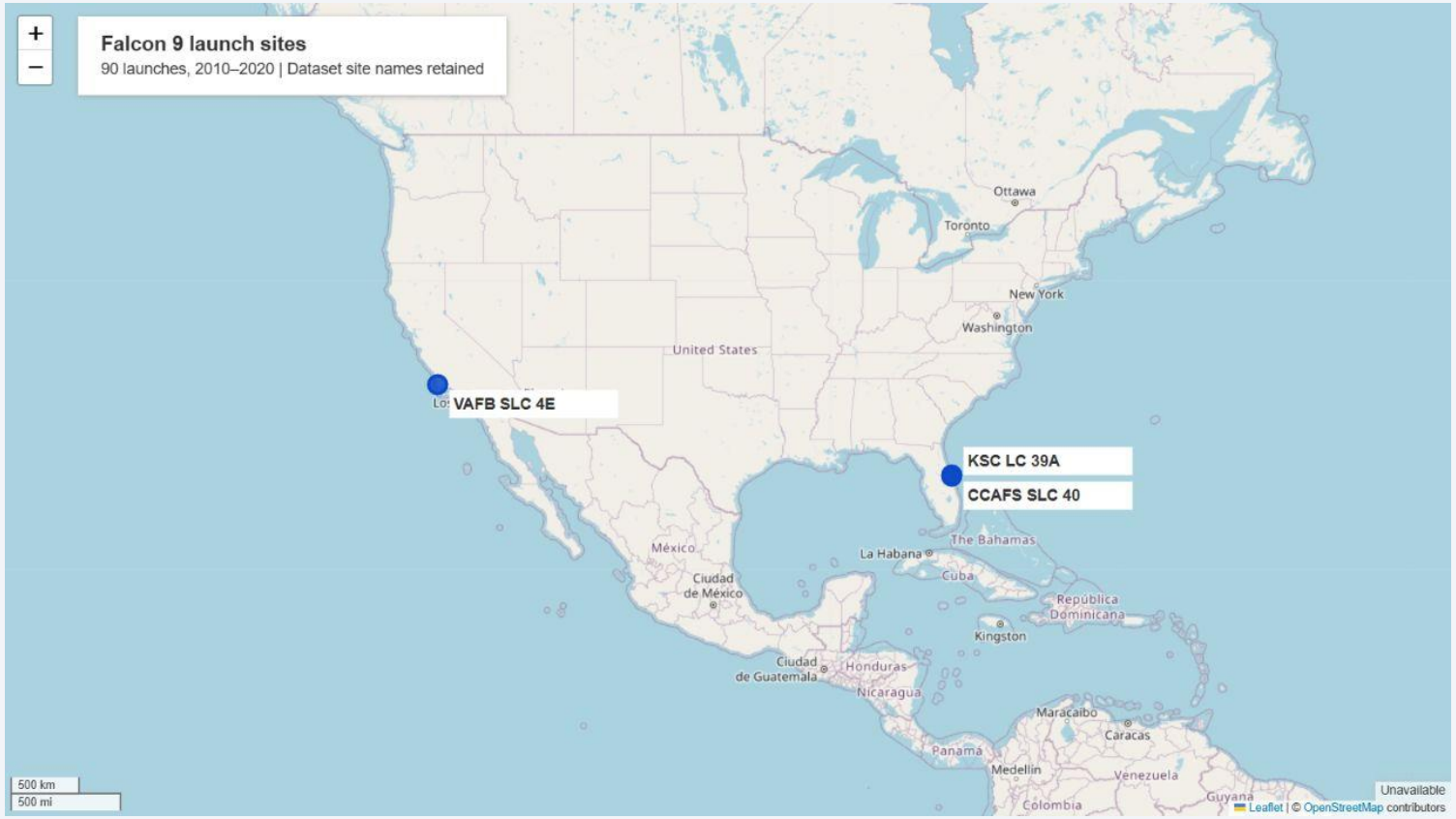
Landing Outcome Ranking, 2010–2017

Landing outcome	Count
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

Inclusive period: 4 June 2010 to 20 March 2017. No attempt is most frequent (10). Drone ship successes and failures tie at five.


Section 3
Launch Sites
Proximities Analysis

Launch Sites on the US Coasts

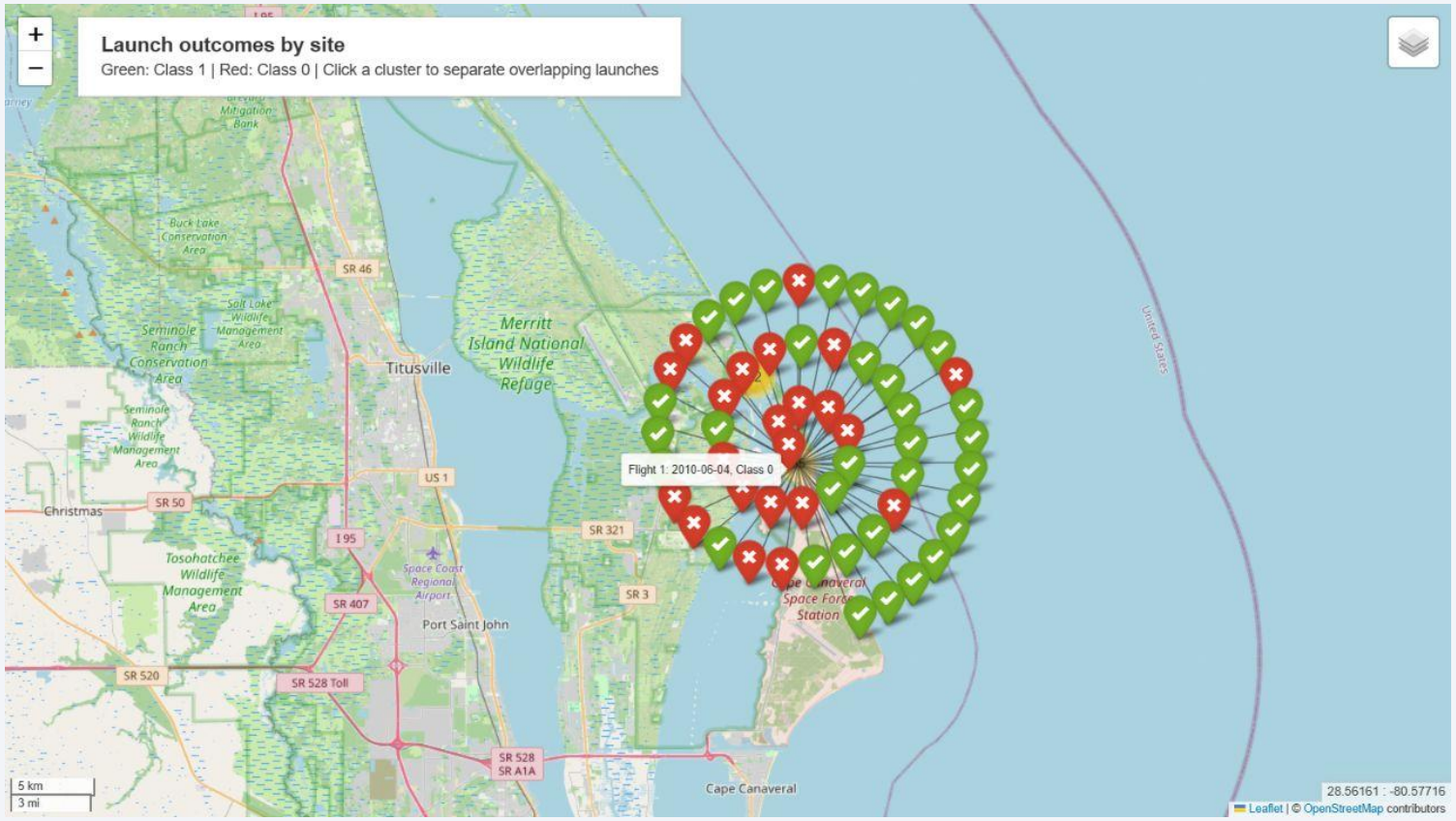


Florida
CCAFS SLC 40
KSC LC 39A

California
VAFB SLC 4E

All three facilities are coastal. The two Florida pads lie close together. Location alone does not explain landing performance.

Launch Outcomes at CCAFS SLC 40



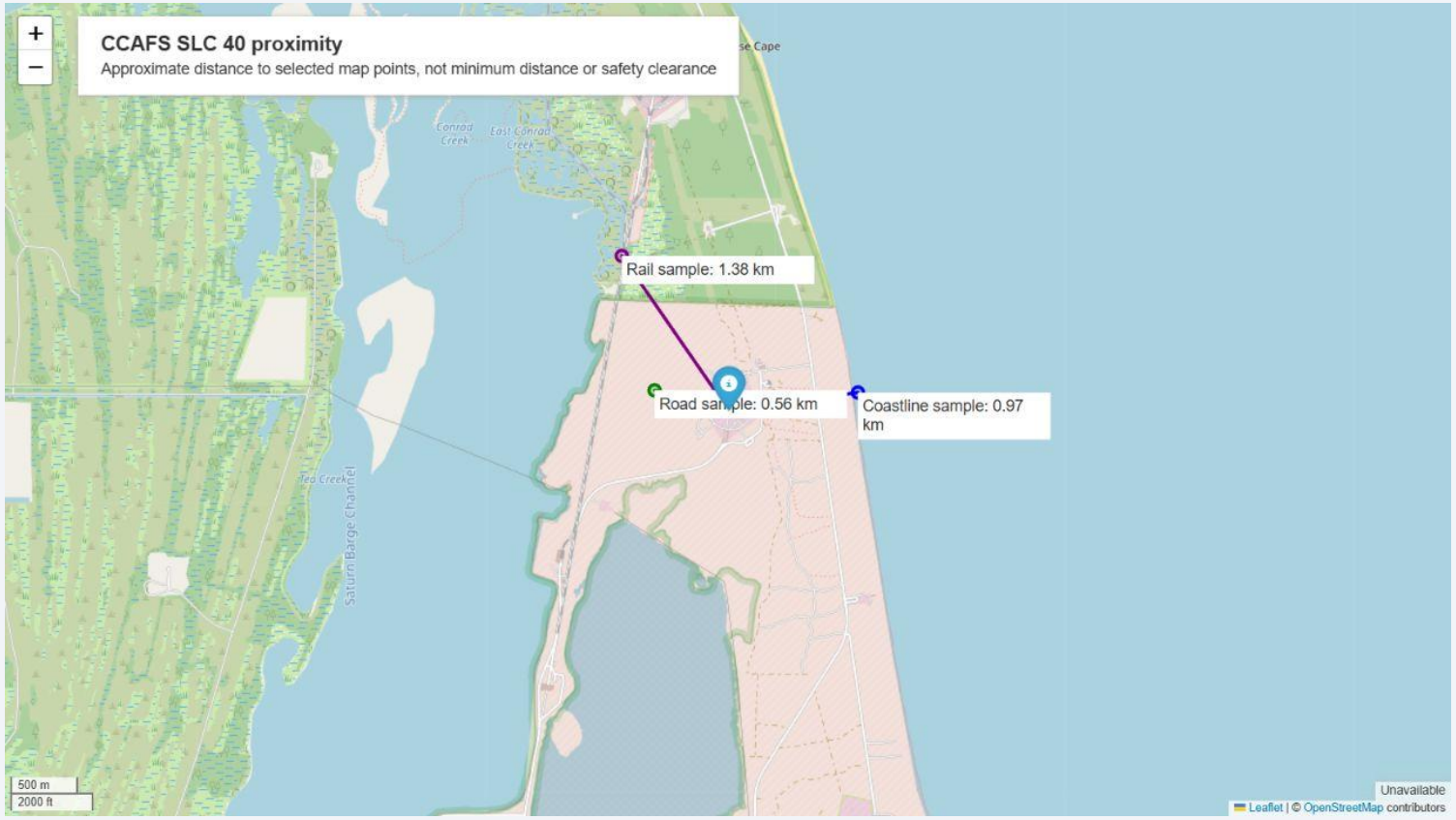
55 launches

33 Class 1
22 Class 0

60.0% success

Expanded markers separate overlapping records around the same launch site. Their displayed offsets are not actual launch locations.

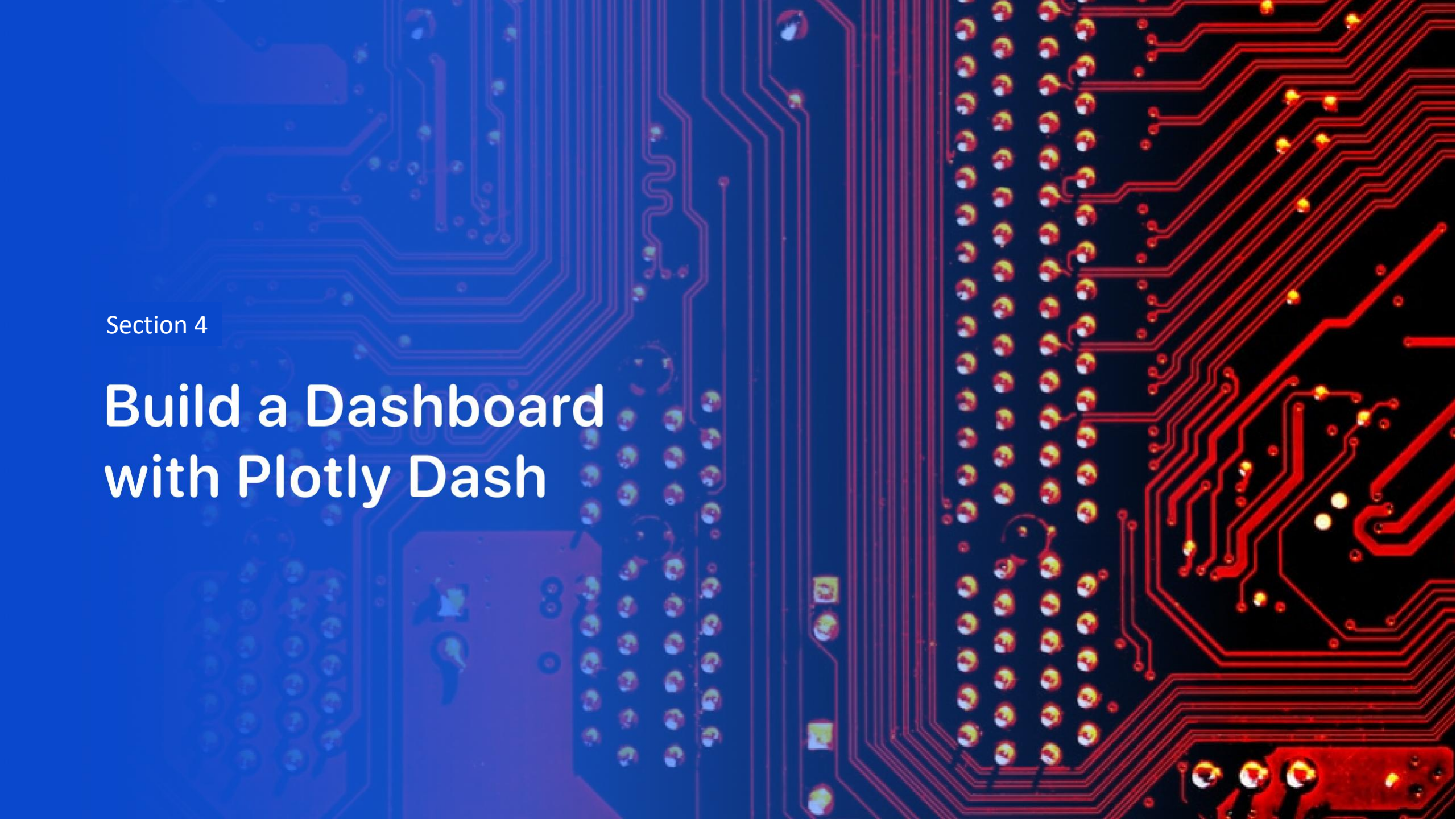
CCAFS SLC 40 and Nearby Features



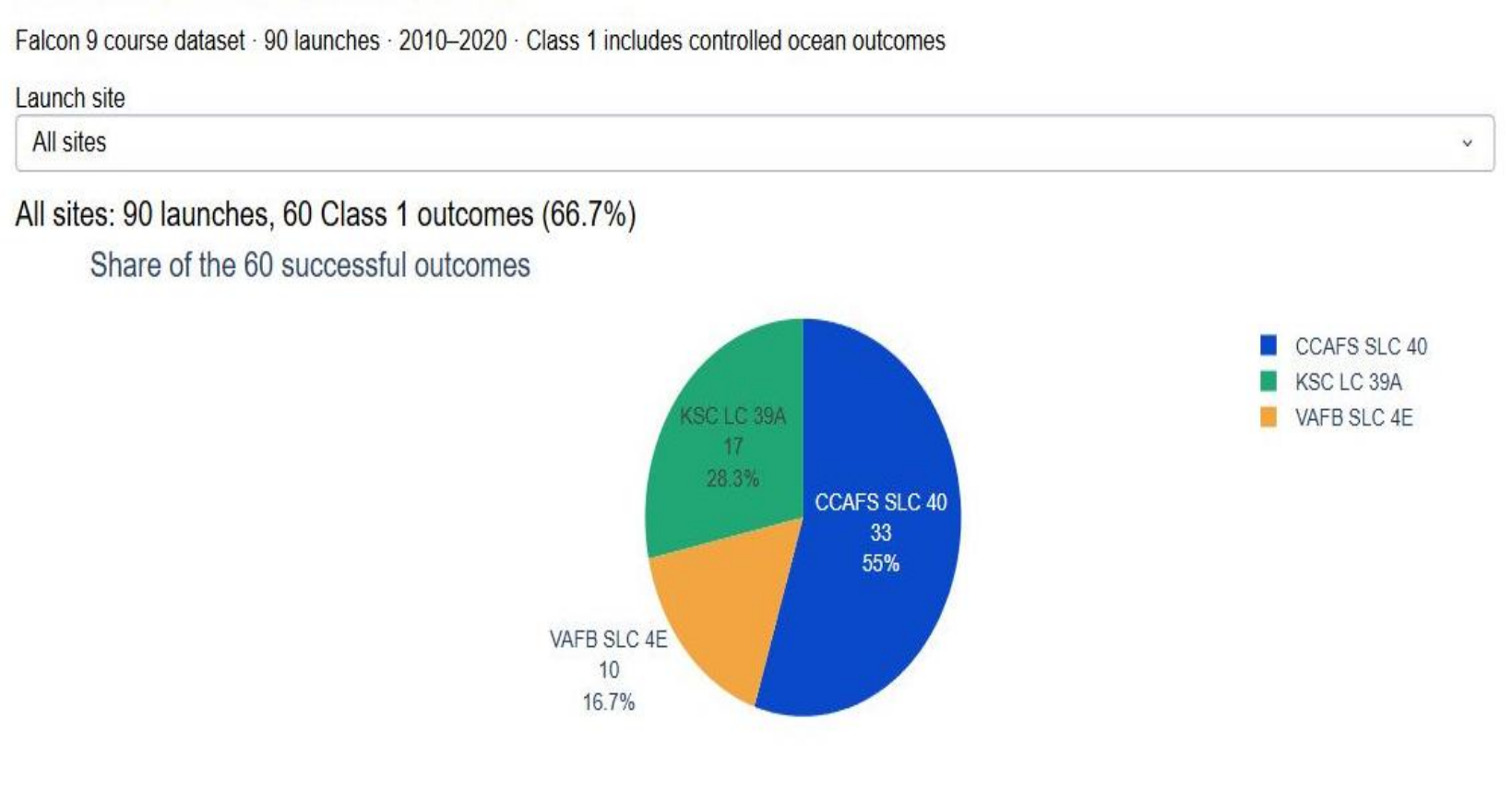
Selected points

Coast: 0.97 km
Road: 0.56 km
Rail: 1.38 km

Approximate straight-line distances to selected map points. These are not surveyed minimum distances or safety clearances.


Section 4
Build a Dashboard with Plotly Dash

Dashboard: Successful Outcomes by Site



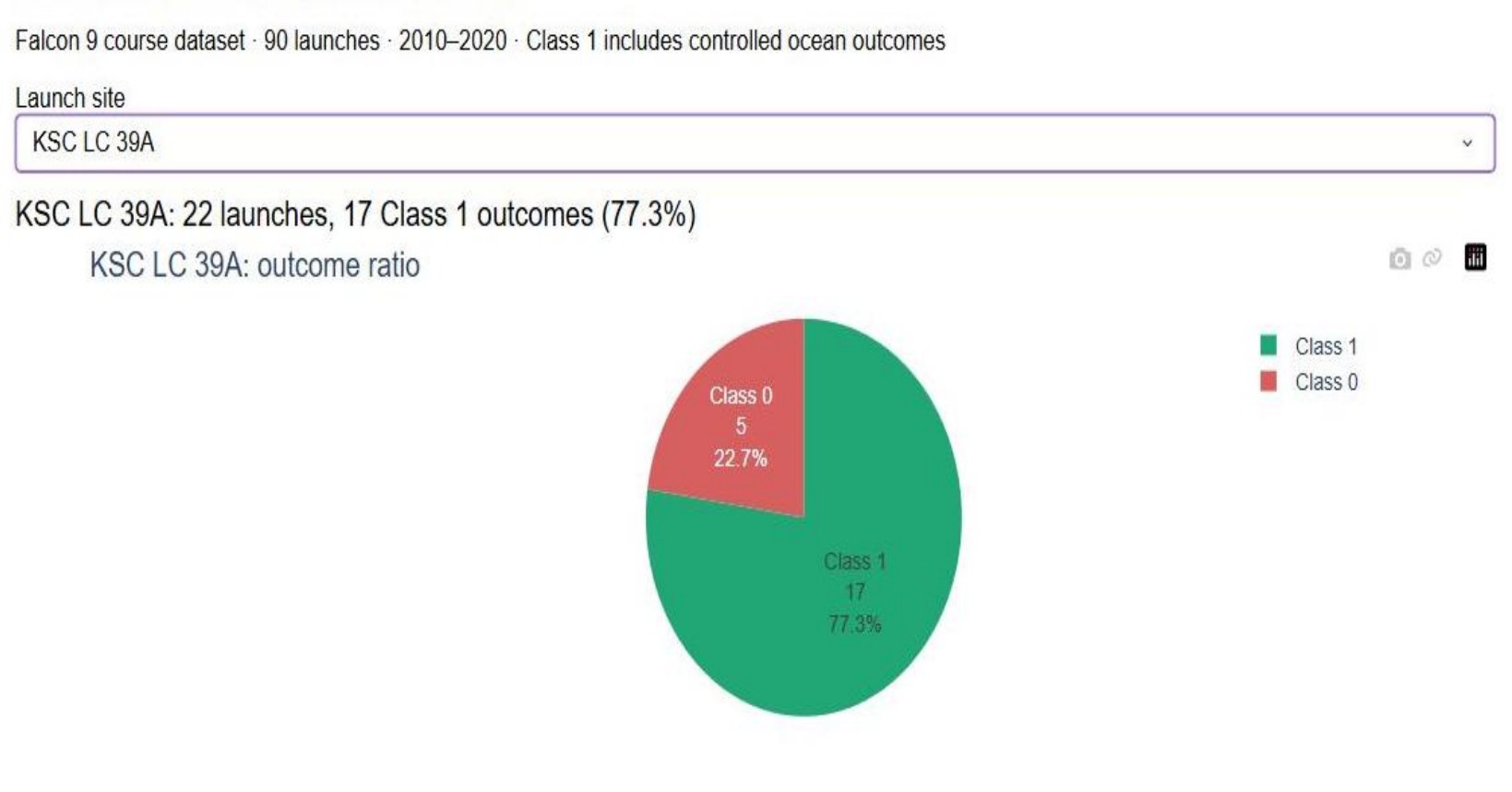
60 successes

CCAFS: 33
KSC: 17
VAFB: 10

CCAFS contributes 55% of successes.

The pie compares successful-outcome counts. A larger slice does not imply a higher success rate.

Dashboard: Highest Observed Site Rate



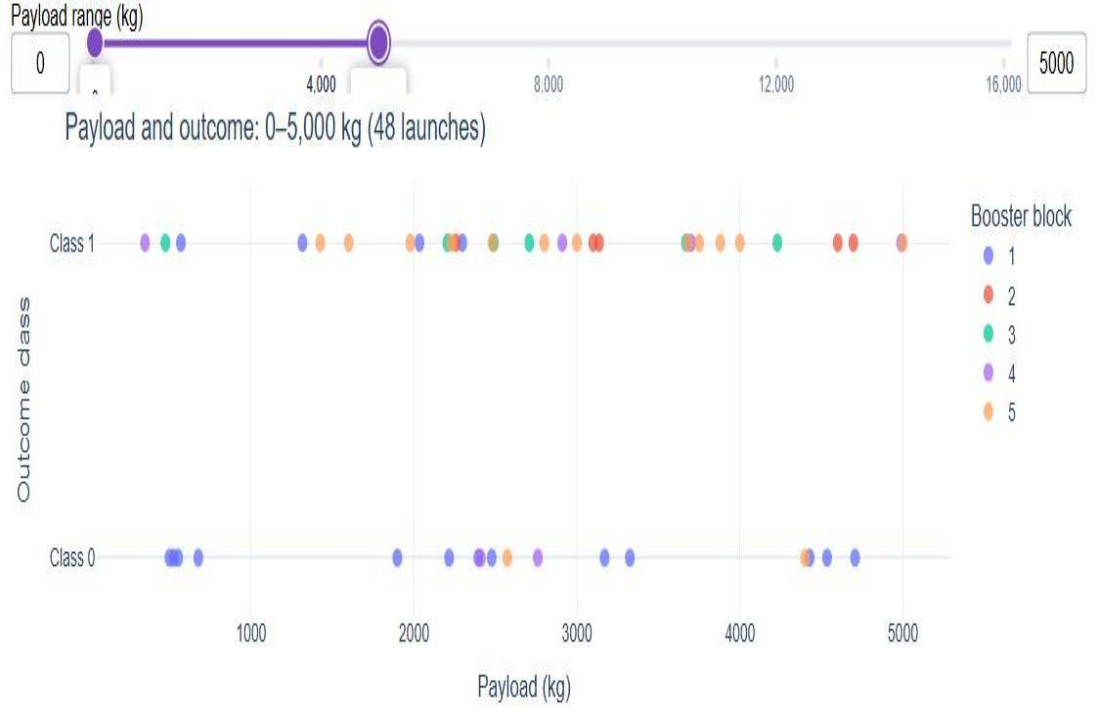
KSC LC 39A

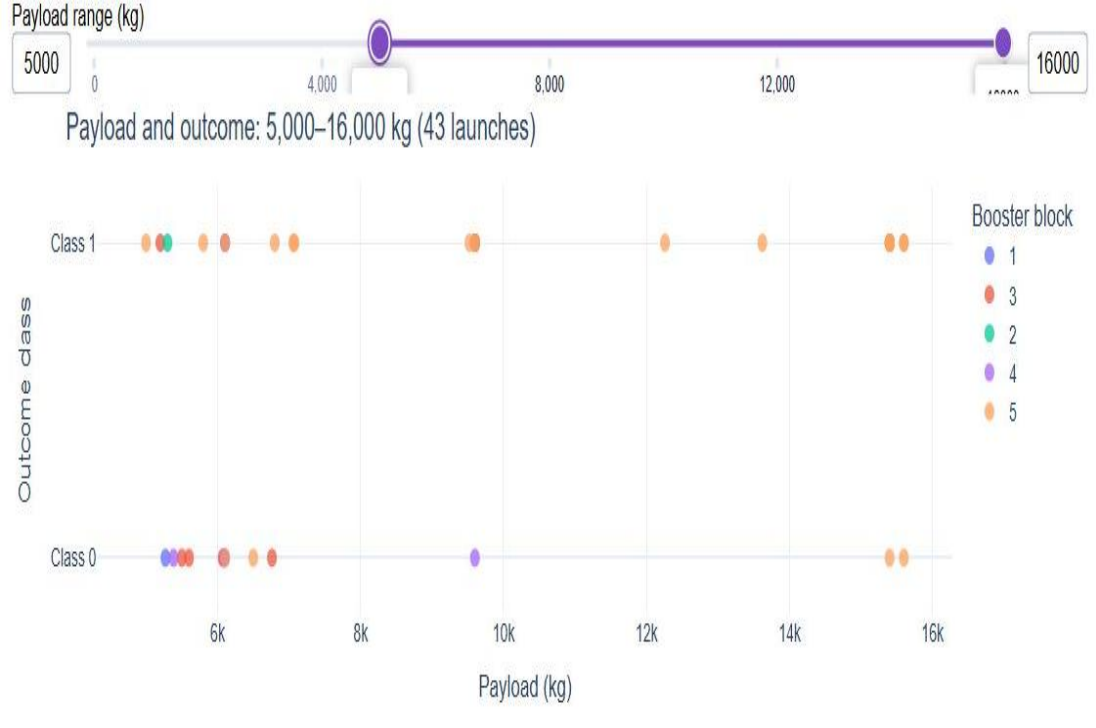
17 successes
5 class 0

77.3% success
22 launches

KSC narrowly exceeds VAFB's 76.9% (10/13). These small samples do not establish a reliable ranking of site quality.

Dashboard: Payload Range Filters





Two slider selections isolate different payload bands. Booster block and flight date also vary, so a payload association is not a causal effect.

Appendix and Reproducible Sources

Public repository

https://github.com/nicholas-yap/spacex-data-science-capstone

notebooks/	Collection, scraping, wrangling, EDA, SQL, maps and classification
data/	Historical course snapshots and scraped records
maps/	Interactive HTML maps
spacex_dashboard.py	Runnable dashboard on port 8051

Thank you!



Nicholas Yap